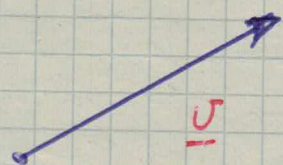


VECTOR COMPUTATIONS

1/

DEF E^3 is the standard Euclidean 3 dimensional space.

Given a segment and a direction the oriented segment is a vector in E^3



The length of $\underline{u}$ is the modulus of the vector, and denoted as $|\underline{u}|$

E^3 is a vector space since

$\forall \underline{u}_1, \underline{u}_2 \in E^3, \forall \alpha, \beta \in \mathbb{R}$ then

$$\underline{u} = \alpha \underline{u}_1 + \beta \underline{u}_2 \in E^3$$

Consider m vectors $\underline{u}_1 \dots \underline{u}_m$ and m coeff. $\alpha_1 \dots \alpha_m$

$$\underline{u} = \alpha_1 \underline{u}_1 + \dots + \alpha_m \underline{u}_m = \sum_{i=1}^m \alpha_i \underline{u}_i$$

is a linear combination of the vectors

DEF The vectors $\underline{u}_1 \dots \underline{u}_m$ are linearly independent

if $\alpha_1 \underline{u}_1 + \dots + \alpha_m \underline{u}_m = \underline{0}$ is possible

only if $\alpha_1 = \alpha_2 = \dots = \alpha_m = 0$

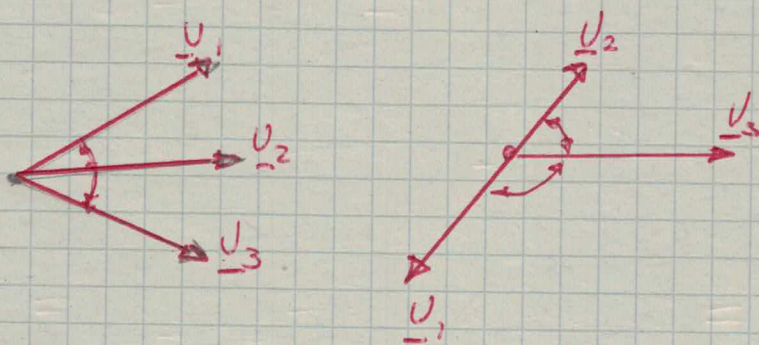
Otherwise the vectors are linearly dependent

NOTE In $\mathbb{R}^3$ two vectors are linearly dependent if they are parallel.

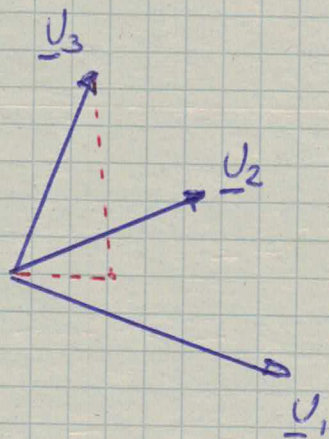
DEF The dimension of a vector space V is the maximum number of linearly independent vectors, denoted as $\dim(V)$

DEF For a given vector space V if $n = \dim(V)$ any set of n linearly independent vectors forms a basis of V

NOTE $\dim(\mathbb{R}^3) = 3$ any set of non coplanar and mutually non co-linear vectors is a basis of $\mathbb{R}^3$



These vectors do not form a basis of $\mathbb{R}^3$



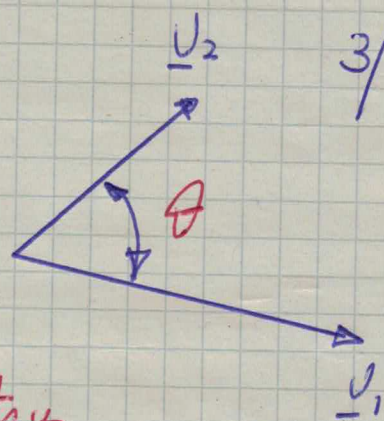
These vectors form a basis of $\mathbb{R}^3$

SCALAR PRODUCT

$$\underline{v}_1, \underline{v}_2, \underline{v}_3 \in \mathbb{E}^3$$

3/

$$\underline{v}_1 \cdot \underline{v}_2 \triangleq |\underline{v}_1| \cdot |\underline{v}_2| \cos \theta \in \mathbb{R}$$



Properties

- $\underline{v}_1 \cdot \underline{v}_2 = \underline{v}_2 \cdot \underline{v}_1$ symmetry
- $\underline{v}_3 \cdot (\alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2) = \alpha_1 (\underline{v}_3 \cdot \underline{v}_1) + \alpha_2 (\underline{v}_3 \cdot \underline{v}_2)$
- $|\underline{v}| = \sqrt{\underline{v} \cdot \underline{v}} \quad (\underline{v} \cdot \underline{v} > 0 \quad \forall \underline{v} \neq \underline{0})$
- $\underline{v}_1 \cdot \underline{v}_2 = 0 \iff \underline{v}_1 \perp \underline{v}_2$

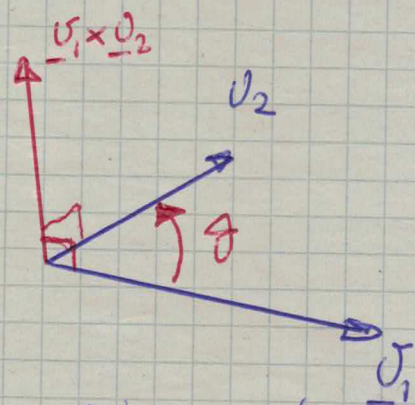
CROSS PRODUCT

$$\underline{v}_1, \underline{v}_2 \in \mathbb{E}^3$$

$$\underline{v}_1 \times \underline{v}_2 \triangleq \underline{v} \in \mathbb{E}^3$$

$$|\underline{v}| = |\underline{v}_1| \cdot |\underline{v}_2| \sin \theta$$

$$\underline{v} \perp \underline{v}_1 \quad \underline{v} \perp \underline{v}_2 \quad \leftarrow \text{right-hand-rule}$$



Properties

- $(\underline{v}_1 \times \underline{v}_2) = -(\underline{v}_2 \times \underline{v}_1)$ anti-symmetry
- $\underline{v}_3 \times (\alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2) = \alpha_1 (\underline{v}_3 \times \underline{v}_1) + \alpha_2 (\underline{v}_3 \times \underline{v}_2)$
- $\underline{v} \times \underline{v} = \underline{0}$
- $\underline{v}_1 \times \underline{v}_2 = \underline{0} \iff \underline{v}_1 \parallel \underline{v}_2$

TRIPLE PRODUCT

$$\underline{v}_1, \underline{v}_2, \underline{v}_3 \in \mathbb{E}^3$$

4/

$$\underline{v}_1 \cdot (\underline{v}_2 \times \underline{v}_3) \in \mathbb{R}$$

Properties

$$\bullet \underline{v}_1 \cdot (\underline{v}_2 \times \underline{v}_3) = \underline{v}_2 \cdot (\underline{v}_3 \times \underline{v}_1) = \underline{v}_3 \cdot (\underline{v}_1 \times \underline{v}_2)$$

(product is preserved through cyclic permutations of the arguments)

$$\bullet \underline{v}_1 \cdot (\underline{v}_2 \times \underline{v}_3) = 0 \iff \underline{v}_1, \underline{v}_2, \underline{v}_3 \text{ are co-planar}$$

$$\bullet \underline{v}_1 \cdot (\underline{v}_2 \times \underline{v}_3) \neq 0 \iff \underline{v}_1, \underline{v}_2, \underline{v}_3 \text{ are linearly indep.}$$

DOUBLE CROSS PRODUCT

$$\underline{v}_1, \underline{v}_2, \underline{v}_3 \in \mathbb{E}^3$$

$$\underline{v}_1 \times (\underline{v}_2 \times \underline{v}_3) = (\underline{v}_1 \cdot \underline{v}_3) \underline{v}_2 - (\underline{v}_1 \cdot \underline{v}_2) \underline{v}_3 \quad (*)$$

$$(\underline{v}_1 \times \underline{v}_2) \times \underline{v}_3 = (\underline{v}_1 \cdot \underline{v}_3) \underline{v}_2 - (\underline{v}_2 \cdot \underline{v}_3) \underline{v}_1 \quad (**)$$

(*) is a linear combination of $\underline{v}_2$ and $\underline{v}_3$

(**) is a linear combination of $\underline{v}_2$ and $\underline{v}_1$

Properties

$$- \underline{v}_1 \times (\underline{v}_2 \times \underline{v}_3) = 0 \quad \text{if } \underline{v}_1 \perp \underline{v}_2 \text{ and } \underline{v}_1 \perp \underline{v}_3$$

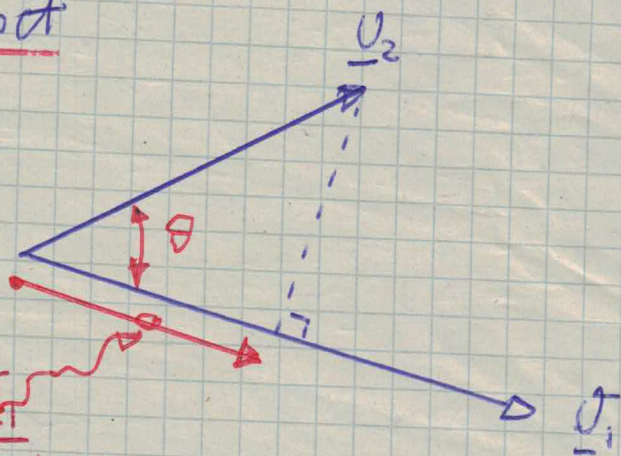
$$- (\underline{v}_1 \times \underline{v}_2) \times \underline{v}_3 = 0 \quad \text{if } \underline{v}_1 \perp \underline{v}_3 \text{ and } \underline{v}_2 \perp \underline{v}_3$$

DEF $\underline{u}$ is a unit vector if $|\underline{u}| = 1$ 5/

Given a generic vector $\underline{v} \neq \underline{0}$ then $\frac{\underline{v}}{|\underline{v}|} \triangleq \frac{1}{|\underline{v}|} \underline{v}$ is a unit vector [OK]

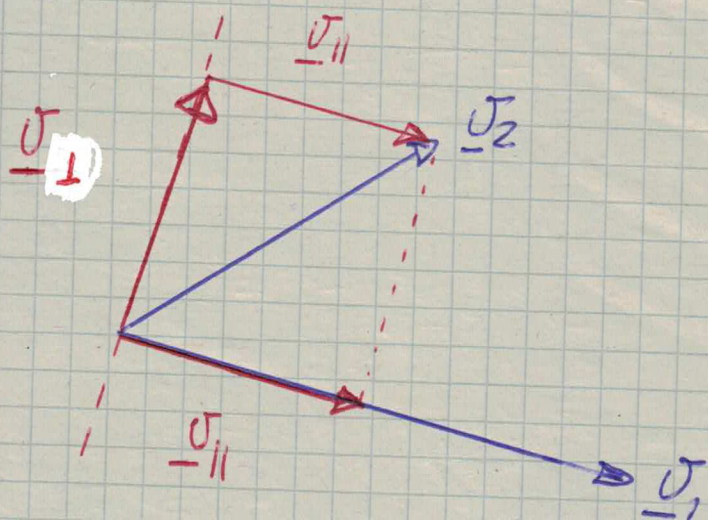
Geometric interpretation of vector operations

Scalar product



$$\underline{v}_{11} = \left(\underline{v}_2 \cdot \frac{\underline{v}_1}{|\underline{v}_1|} \right) \frac{\underline{v}_1}{|\underline{v}_1|}$$

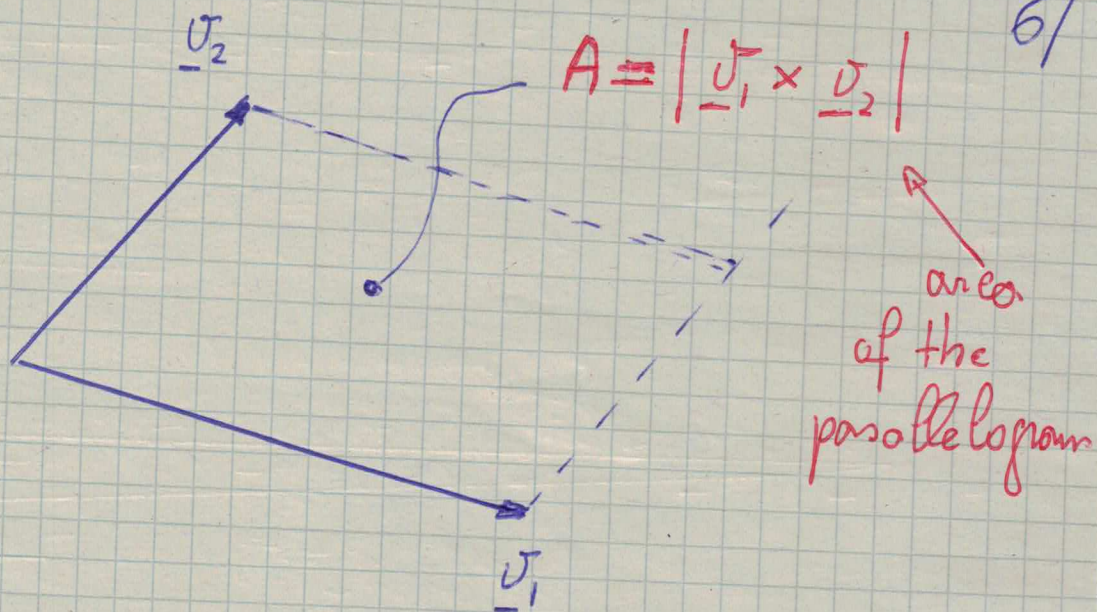
orthogonal projection of $\underline{v}_2$ along the direction of $\underline{v}_1$



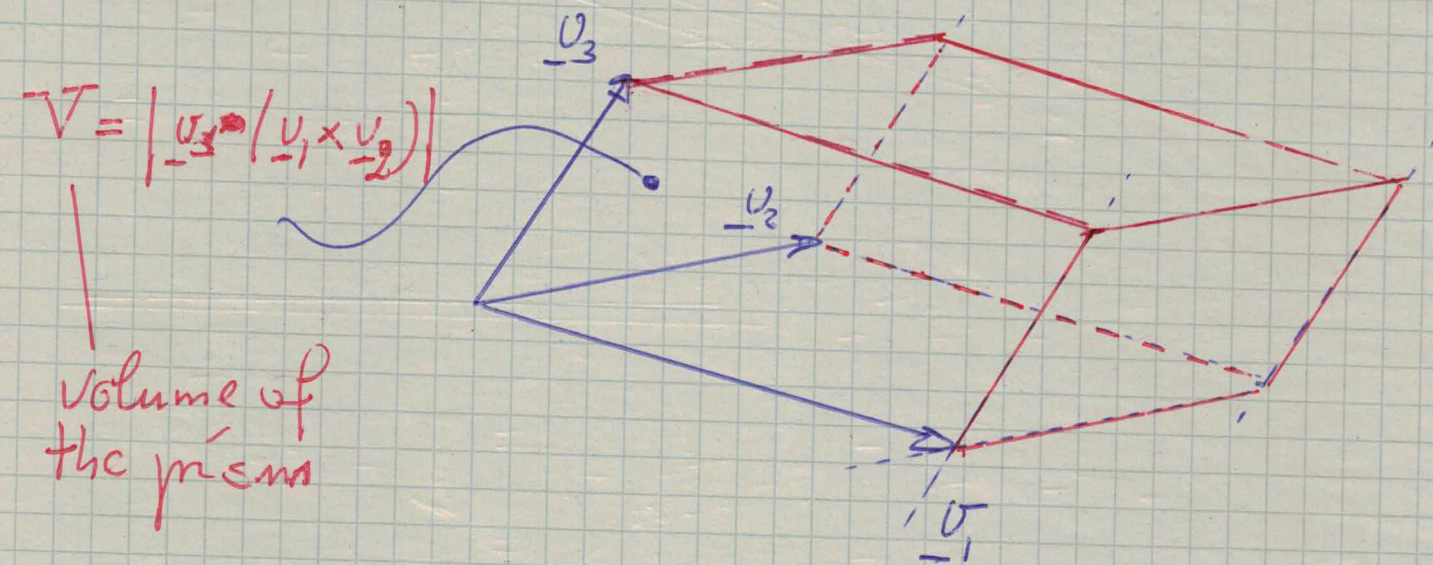
$$\underline{v}_{\perp} = \underline{v}_2 - \underline{v}_{11} = \underline{v}_2 - \left(\underline{v}_2 \cdot \frac{\underline{v}_1}{|\underline{v}_1|} \right) \frac{\underline{v}_1}{|\underline{v}_1|}$$

orthogonal component of $\underline{v}_2$ with respect to $\underline{v}_1$

Cross product



Triple product



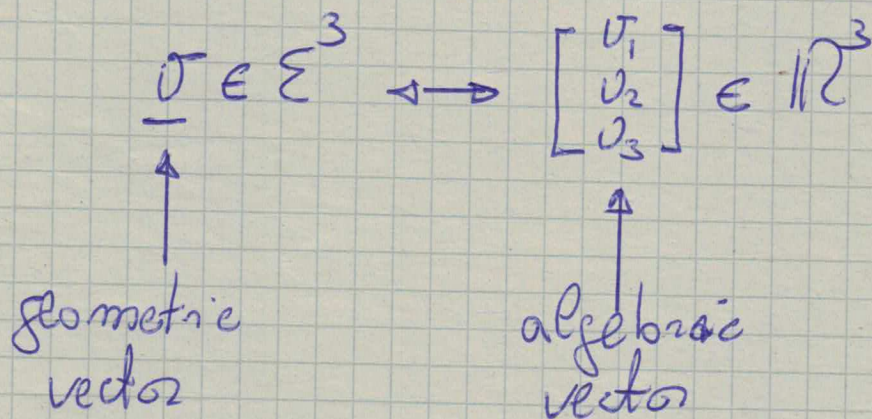
DEF $\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$ is an orthonormal right-handed basis of $\mathbb{E}^3$ if

- $|\underline{e}_i| = 1 \quad \forall i$
- $\underline{e}_i \cdot \underline{e}_j = 0 \quad \forall i, j$
- $(\underline{e}_1 \times \underline{e}_2) \cdot \underline{e}_3 = +1 \quad \leftarrow \text{(right-handed)}$

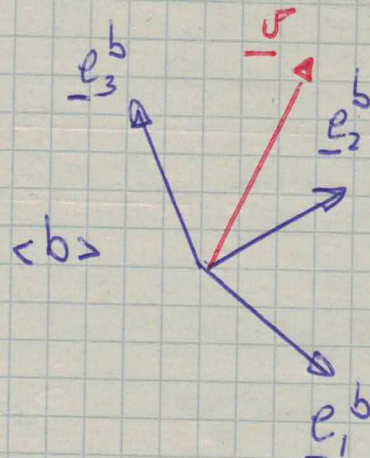
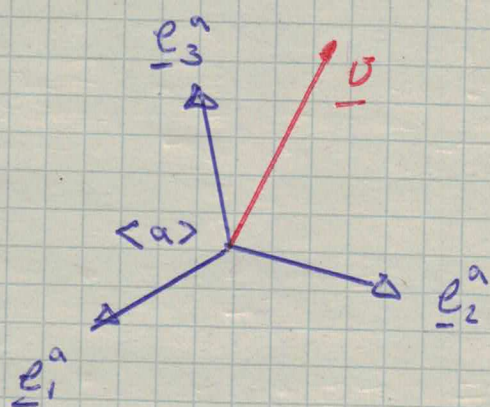
Any vector $\underline{v} \in \mathbb{E}^3$ can be expressed as a unique linear combination of $\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$

$$\underline{v} = \sum_{i=1}^3 v_i \underline{e}_i = \sum_{i=1}^3 (\underline{v} \cdot \underline{e}_i) \underline{e}_i$$

↑
components
of $\underline{v}$ along $\underline{e}_i$



Since there exist infinite many orthonormal basis, for each $\underline{v} \in \mathbb{E}^3$ there exist different algebraic vectors ^{each} related to a single basis



$$\underline{v} \mapsto \begin{bmatrix} {}^a v_1 \\ {}^a v_2 \\ {}^a v_3 \end{bmatrix} \triangleq \underline{{}^a v}$$

$$\underline{v} \mapsto \begin{bmatrix} {}^b v_1 \\ {}^b v_2 \\ {}^b v_3 \end{bmatrix} \triangleq \underline{{}^b v}$$

in general $\underline{{}^a v} \neq \underline{{}^b v}$ for the same $\underline{v}$

COORDINATE TRANSFORMATIONS

Consider two orthonormal bases $\langle a \rangle, \langle b \rangle$

$$\underline{e}_i^b = \sum_{j=1}^3 (\underline{e}_i^b \cdot \underline{e}_j^a) \underline{e}_j^a \quad \forall i$$

↑
representation of $\langle b \rangle$
with respect to $\langle a \rangle$

$${}^a(\underline{e}_i^b) = \begin{bmatrix} (\underline{e}_i^b \cdot \underline{e}_1^a) \\ (\underline{e}_i^b \cdot \underline{e}_2^a) \\ (\underline{e}_i^b \cdot \underline{e}_3^a) \end{bmatrix} \quad \forall i$$

$${}^a_b R \triangleq \begin{bmatrix} (\underline{e}_1^b \cdot \underline{e}_1^a) & (\underline{e}_2^b \cdot \underline{e}_1^a) & (\underline{e}_3^b \cdot \underline{e}_1^a) \\ (\underline{e}_1^b \cdot \underline{e}_2^a) & (\underline{e}_2^b \cdot \underline{e}_2^a) & (\underline{e}_3^b \cdot \underline{e}_2^a) \\ (\underline{e}_1^b \cdot \underline{e}_3^a) & (\underline{e}_2^b \cdot \underline{e}_3^a) & (\underline{e}_3^b \cdot \underline{e}_3^a) \end{bmatrix} = \begin{bmatrix} {}^a(\underline{e}_1^b) & {}^a(\underline{e}_2^b) & {}^a(\underline{e}_3^b) \end{bmatrix}$$

$${}^a R \in \mathbb{R}^{3 \times 3}$$

rotation matrix

Properties

$$\bullet ({}^a R)^T ({}^a R) = \mathbb{I}_3 \quad \Leftrightarrow ({}^a R)^T = ({}^a R)^{-1}$$

$$\bullet ({}^a R)^T = {}^b R$$

$$\bullet {}^a R = {}^a R \cdot {}^b R$$

← sequence of rotations

NOTE A rotation matrix R can be used in different ways

i) ${}^b \underline{v}$ is the representation of $\underline{v}$ in frame $\langle b \rangle$ then

$${}^a \underline{v} = R {}^b \underline{v}$$

← representation of $\underline{v}$ in frame $\langle a \rangle$ given the representation in $\langle b \rangle$

ii) ${}^c \underline{v}$ is the representation of $\underline{v}$ in frame $\langle c \rangle$ (*) then

$${}^c \underline{v}_R = R {}^c \underline{v}$$

← representation (in $\langle c \rangle$) of $\underline{v}$ rotated as specified by R (more details required...)

$\underline{v}$ rotated as specified by R

(*) generic frame